

TABLEAU ASSIGNMENT:

THE IMPACT OF COVID-19 PANDEMIC ON ROAD SAFETY

Group 35

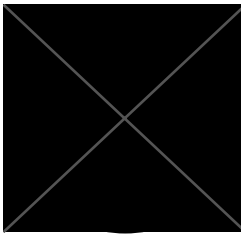
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MOTIVATION



Understanding COVID-19's impact on road safety in Finland

- The pandemic significantly altered daily life, including how and when people travel.
- It's unclear whether these changes led to safer or more dangerous roads.
- A data-driven analysis is crucial to separate real trends from assumptions.



Targeting long-term traffic safety improvements

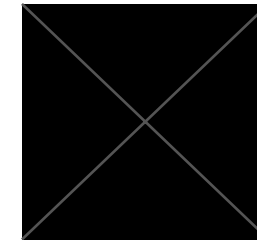
- Identifying accident patterns during the pandemic can reveal hidden risks or protective factors.
- These insights can inform permanent safety measures beyond COVID-19.
- Helps shift focus from reactive to proactive road safety strategies.



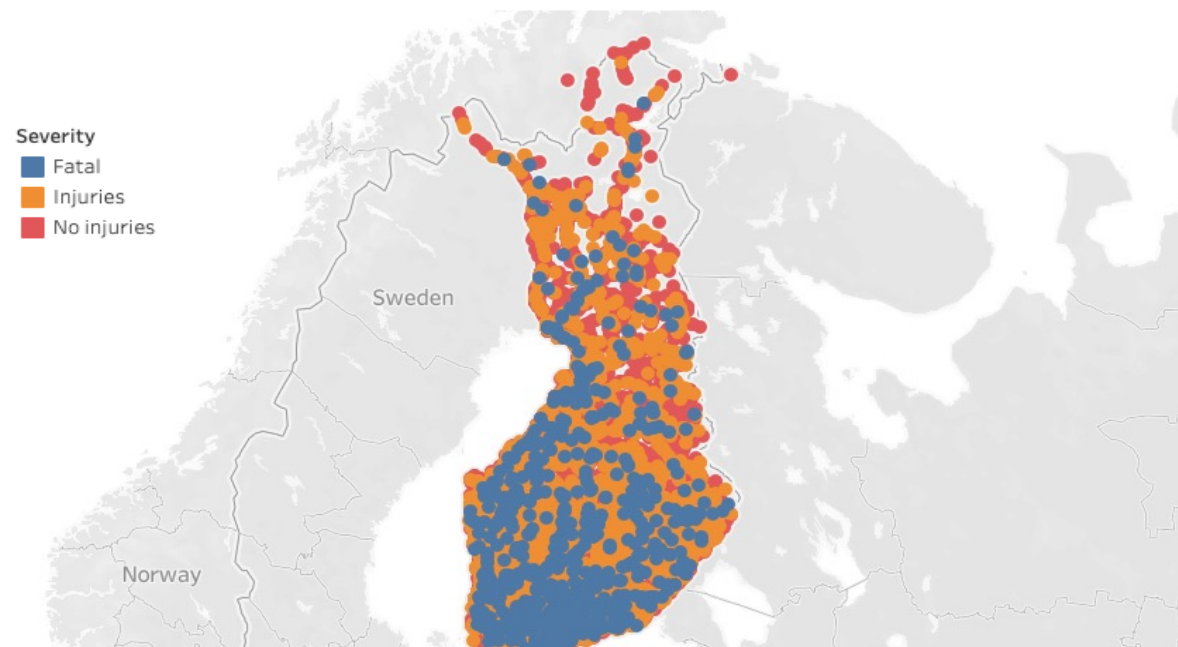
Supporting evidence-based policy and planning

- Authorities need reliable data to make informed decisions about infrastructure and traffic policies.
- Data-driven findings can support smarter urban mobility planning and emergency readiness.
- Ensures future strategies are guided by trends, not just temporary observations.

GEOGRAPHIC DISTRIBUTION OF TRAFFIC ACCIDENTS



Finnish highways

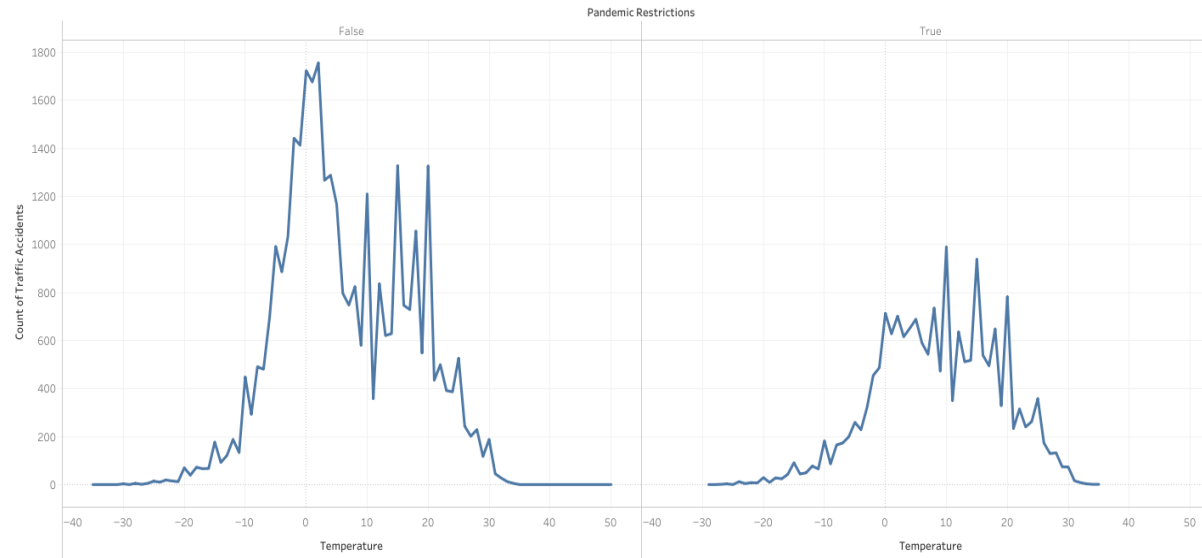


- The most serious accidents are concentrated along highways.
- In sparsely populated areas, the lack of traffic supervision likely plays a major role.
- In the north, road conditions may be a particularly significant factor.

SHIFTING PATTERNS IN ACCIDENT PREVALENCE DURING THE PANDEMIC

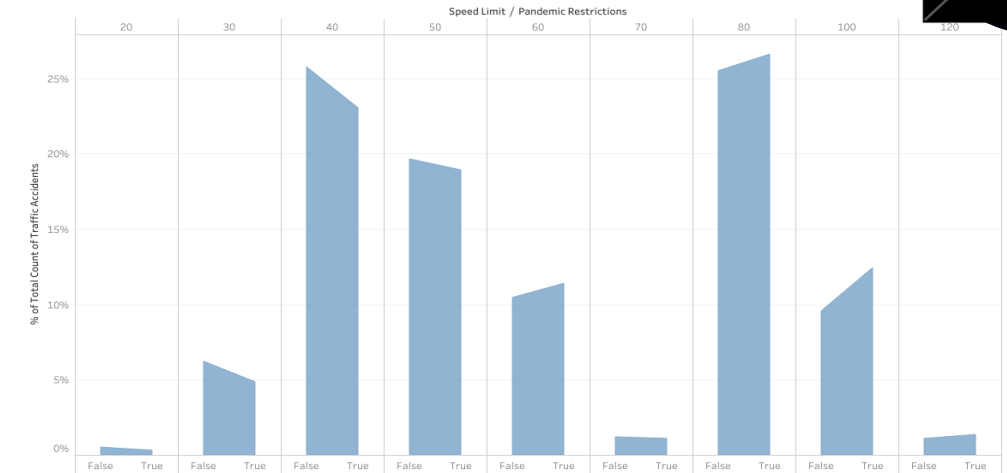
Pandemic restrictions disrupted the seasonal pattern of temperature-related accidents

Traffic Accidents Seasonality



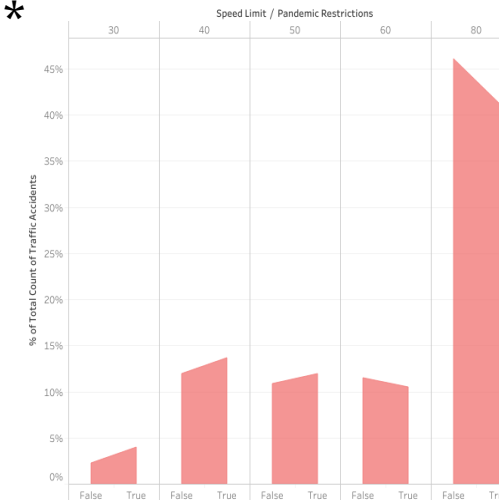
*Excluding speed limits 70,100,120 from visualization for greater clarity

Distribution of Traffic Accidents by Speed Limit by Pandemic Restrictions



Distribution of Lethal Traffic Accidents by Speed Limit by Pandemic Restrictions

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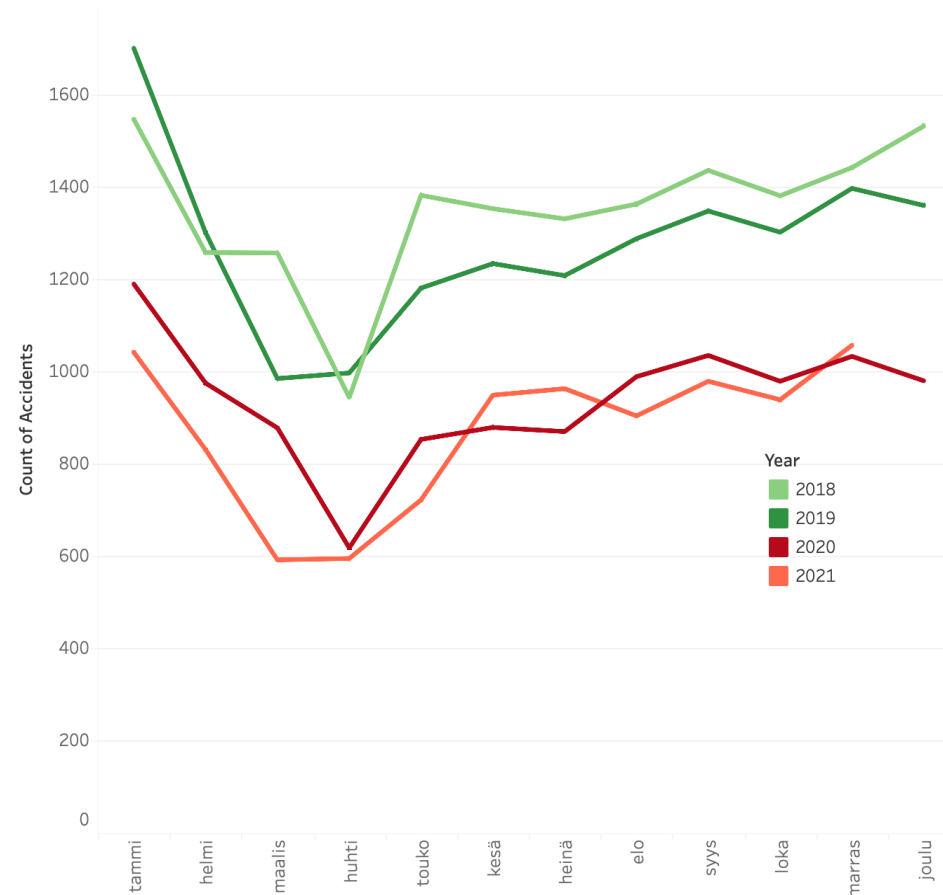
The relative prevalence of accidents shifts around the 50 km/h speed limit, indicating a pivot in distribution trends (5,08 % prevalence distribution shift from instances less or equal to 50 km/h)

- <--However, the distribution trend for fatal accidents follows an inverse relationship (4,58 % prevalence distribution shift from instances greater or equal to 60 km/h)

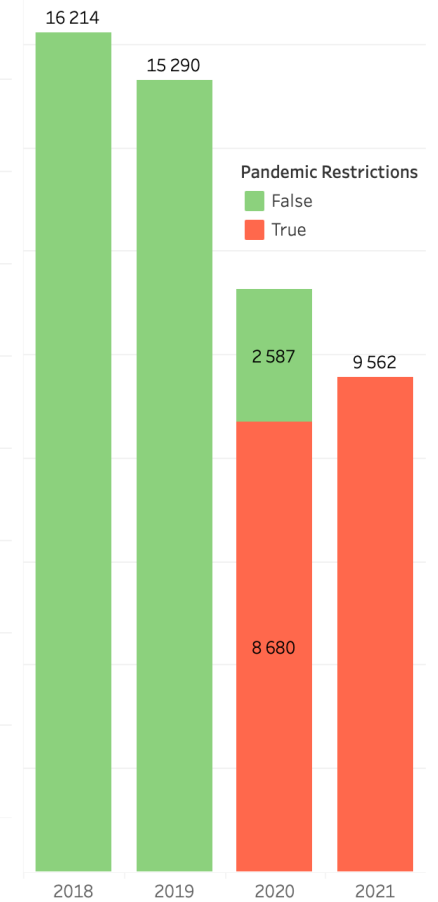
ACCIDENT TREND DEVELOPMENT

- **Yearly traffic accidents decreased by 40% (over 6000 yearly accidents) from 2018 to 2021 and by 37% from 2019 to 2021**
 - Some of the decrease may be explained by a general decreasing trend in traffic accidents
- All years show a seasonal trend.
- In 2020, a clear drop occurred in March–April, likely due to the onset of the COVID-19 pandemic.
- Accidents were consistently highest in 2019, especially in January and February.
- 2021 had the lowest monthly accident counts overall, remaining below all previous years throughout.

Monthly accident trend by year

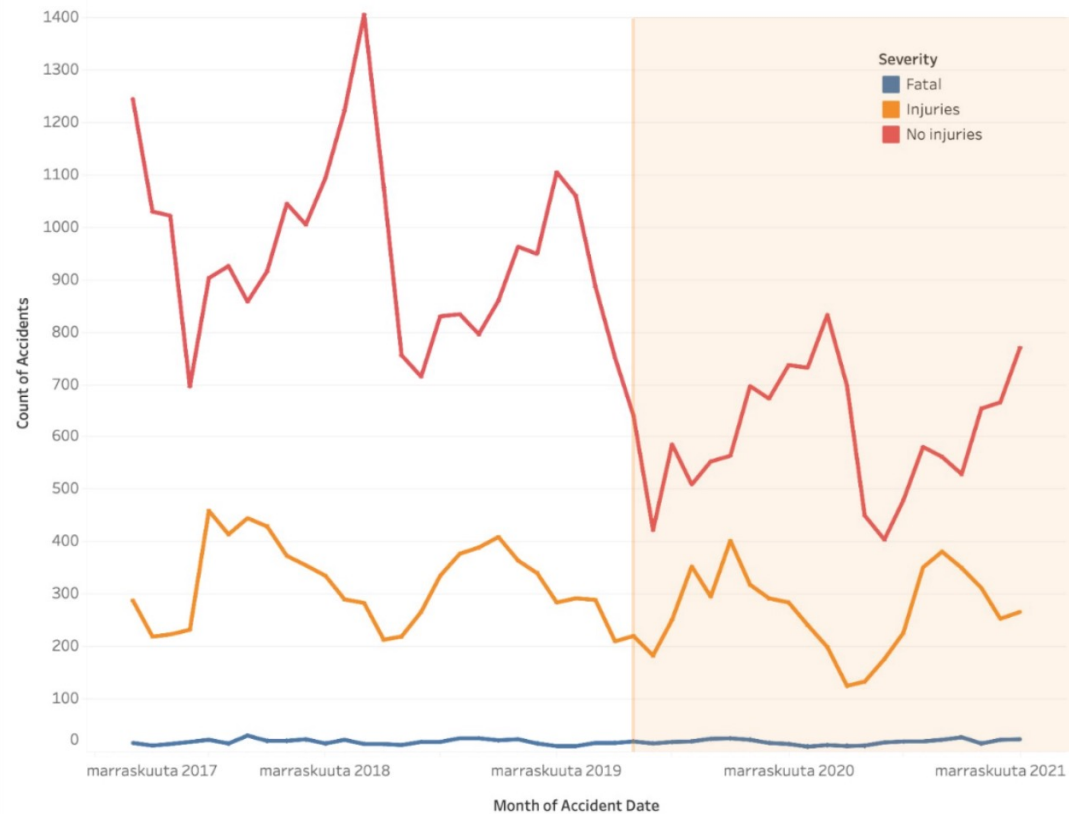


Yearly total accidents

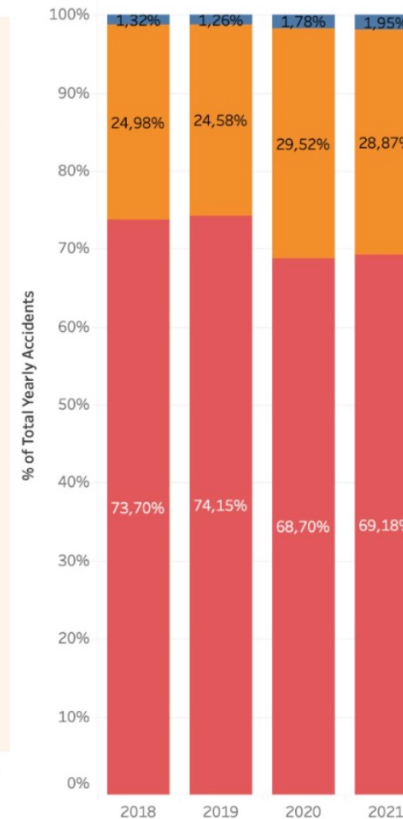


CHANGE IN ACCIDENT SEVERITY OVER THE PANDEMIC

Severity trend



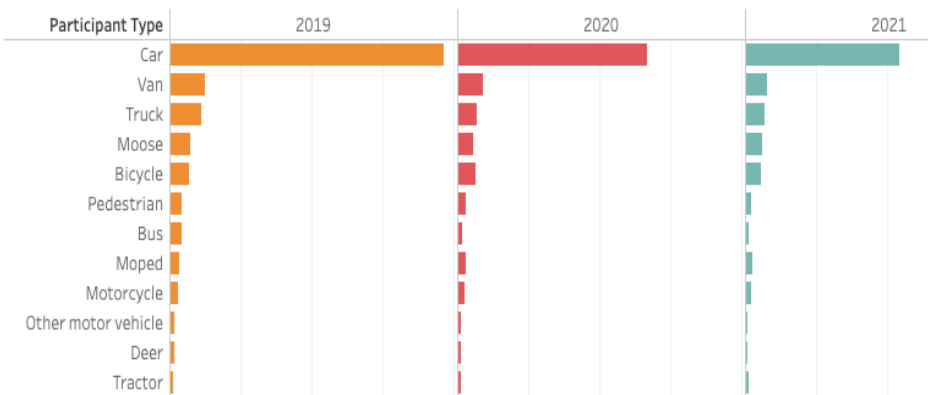
Accident severity by year



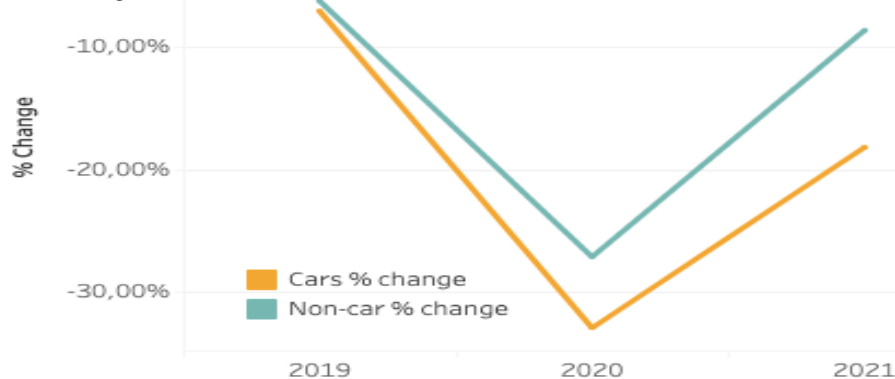
- Although the general number of accidents significantly decreased during the pandemic, **the relative proportion of severe accidents increased**
- From 2019 to 2020 **the relative proportion of fatal accidents increased by 55% and accidents leading to injuries by 18%**
- The relative proportion of accidents leading to no injuries decreased by 8,82% from 2019 to 2020
- Note: The decreasing trend began already in January before pandemic restrictions held

CHANGE IN ACCIDENT PARTICIPANT TYPE OVER THE PANDEMIC

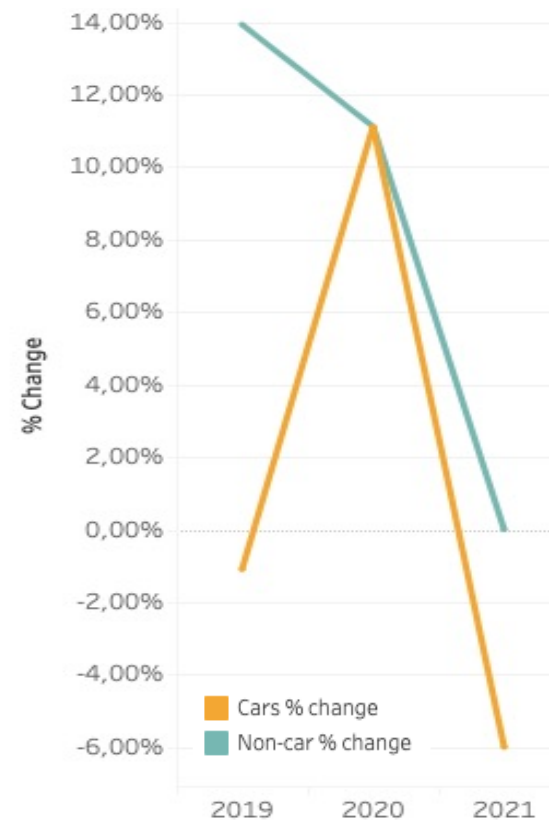
Participant types in all accidents



% Change of all accidents



% Change of fatal accidents



Fatal accidents increased more steeply for cars than for non-car participants in 2020, despite an overall drop in traffic volumes.

This may reflect **riskier driving behavior** in emptier roads, such as speeding or impaired driving during the pandemic.

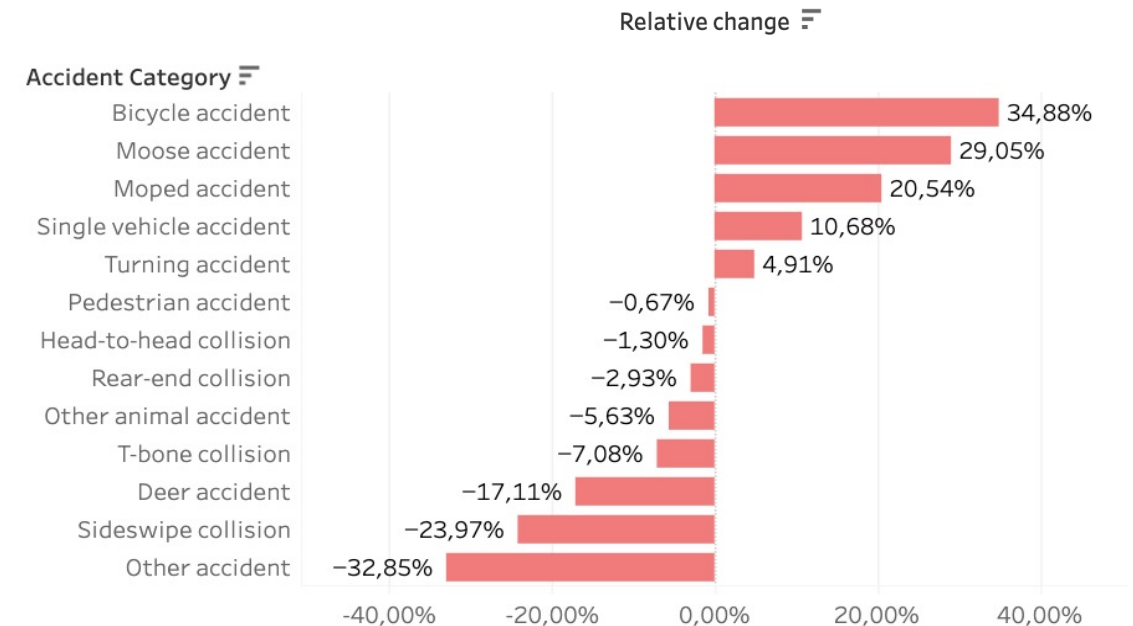
In 2021, fatal car accidents dropped significantly, while non-car fatalities remained relatively stable

*Non-car includes all participant types except cars

CHANGES IN ACCIDENT CATEGORIES DURING COVID RESTRICTIONS

- Moose and bicycle accidents became two of the six most common accident categories during COVID-19 restrictions, compared to before restrictions
- Bicycle, pedestrian, and head-to-head collisions are the three most deadly* accident categories, making the rise in bicycle accidents a potential contributor to the increase in severe accidents

Pandemic Restrictions		Pandemic Restrictions	
True		False	
Accident Category		Accident Category	
Single vehicle accident	26,95%	Single vehicle accident	24,35%
Moose accident	13,15%	Other accident	15,95%
Other accident	10,71%	Rear-end collision	10,57%
Rear-end collision	10,26%	Moose accident	10,19%
T-bone collision	7,87%	T-bone collision	8,47%
Bicycle accident	6,96%	Sideswipe collision	7,01%

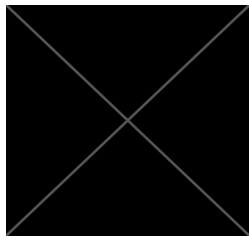


In terms of relative change** during pandemic restrictions:

- The share of bicycle, moose, and moped accidents saw a clear increase
- The share of single-vehicle accidents increased by 10%
- The share of deer accidents, sideswipes, and other accident types experienced a significant decrease during the restriction period

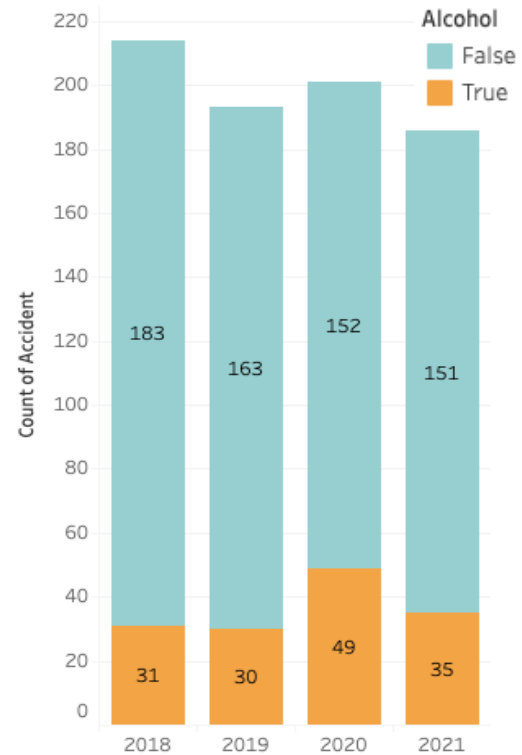
*calculated from all the fatal accidents in the data

**calculation: (accident category % during restrictions / accident category % outside of restrictions) -100%



FATAL ACCIDENTS DURING COVID-19: KEY CONTRIBUTING FACTORS

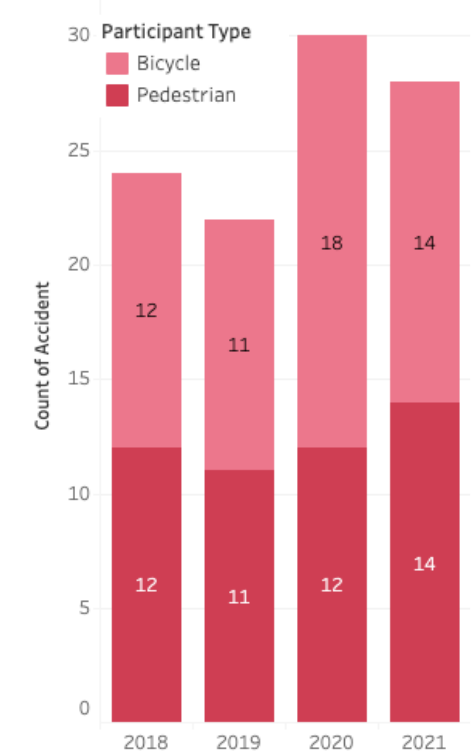
Alcohol-Related and Other Fatal Accidents



Fatal Alcohol-Related Accidents by Hour and Year



Fatal Accidents Involving Cyclists and Pedestrians by Year



KEY INSIGHTS



Accident Trends & Severity

Data:

Total accidents dropped by ~**40%** (approximately 6000 fewer) from 2018–2021, yet fatal accidents increased proportionally by **54.8%** and severe (non-fatal) collisions by **18%** from 2019 to 2020, while non-severe accidents declined by **8.82%**.

Insight:

Fewer collisions occurred overall, but the **remaining crashes were significantly more severe**—suggesting that reduced traffic volumes may have inadvertently encouraged riskier driving behaviors.



Shifts in Participant & Accident Types

Data:

Car accidents fell sharply (a **32.92%** reduction from 2019 to 2020), while non-car incidents involving cyclists, pedestrians, and mopeds either **declined less or increased proportionally**; additionally, incidents like moose and bicycle collisions became more frequent during COVID-19 restrictions.

Insight:

Pandemic-related lifestyle changes (e.g., increased outdoor activity, avoidance of public transport) shifted the road user mix toward more **vulnerable groups** and altered accident types, contributing to **higher injury severity**.



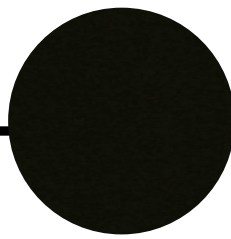
Risky Driving Behaviors & Alcohol

Data:

Emptier roads during the pandemic saw an increase in high-speed and **risk-prone driving**, while remote work and changes in social routines led to a notable shift in **alcohol consumption in nighttime and weekends**.

Insight:

These behavioral shifts—combined with less congested roads—resulted in a **disproportionate rise in severe and fatal accidents**, even as overall collision counts dropped.



LONG-TERM RECOMMENDATIONS: HOW TO REDUCE THE NUMBER OF ACCIDENTS OVERALL

- **Target high-risk time periods:**

Focus enforcement and awareness campaigns on weekends and nighttime, when fatal and alcohol-related accidents are more prevalent.

- **Road-Specific Enforcement & Infrastructure Improvements:**

Use targeted enforcement (e.g., digital speed monitoring) on low traffic & rural roads to maximize limited resources and consider basic infrastructure upgrades on high-risk roads and highways.

- **Strengthen Protection for Vulnerable Road Users:**

Improve infrastructure, control speeds, and promote safety awareness for drivers and high-risk collision participants, such as pedestrians, cyclists, and other non-car travelers.

- **Monitor & Reinforce Safe Driving Behaviors:**

Regularly analyze traffic data and coordinate among agencies to adapt policies swiftly, while launching education campaigns that foster responsible driving habits long after COVID.